

DIABETES RAG KNOWLEDGE BASE

Structured educational reference for retrieval-augmented generation

Educational use only. Not a substitute for professional medical advice, diagnosis, or treatment. No individualized diagnosis, prescriptions, insulin dosing, or medication adjustments. For urgent symptoms, seek immediate medical help.

RAG design notes

This document uses stable section IDs, explicit topics, concise answer blocks, and question-oriented phrasing. For best retrieval, split on DM section headings first, preserve section_id/topic/source/page metadata, and keep the evaluation question bank separate from knowledge chunks.

Grounding rule: Answer only from retrieved passages. If evidence is missing, say so. Generate citations in application code from retrieved metadata rather than asking the model to invent section/page citations.

DM-001: Definition and core concepts

Topic: Overview **Source:** WHO (2023), ADA (2024)

What is diabetes mellitus?

A chronic metabolic disease characterized by elevated blood glucose. It occurs when the pancreas does not produce enough insulin or when the body cannot effectively use the insulin it produces.

What does insulin do?

Insulin is a hormone that helps regulate blood glucose by facilitating glucose uptake into cells for energy.

Why can chronic hyperglycemia be harmful?

Over time, uncontrolled high blood glucose can damage the heart, blood vessels, eyes, kidneys, and nerves.

DM-002: Types of diabetes

Topic: Classification **Source:** ADA (2024), CDC (2024)

Type 1 diabetes (T1D)

An autoimmune condition in which the immune system attacks and destroys insulin-producing pancreatic beta cells, causing absolute insulin deficiency. It often begins in childhood or adolescence but can occur at any age.

Type 2 diabetes (T2D)

Progressive inadequate insulin secretion, commonly associated with insulin resistance. The supplied reference states it accounts for approximately 90–95% of diabetes cases.

Gestational diabetes (GDM)

Diabetes diagnosed during pregnancy; the reference treats it as a distinct category.

How do T1D and T2D differ?

T1D involves autoimmune beta-cell destruction and absolute insulin deficiency. T2D commonly begins with insulin resistance and may progress to beta-cell dysfunction and insufficient insulin secretion.

DM-003: Causes, risk factors, pathophysiology

Topic: Etiology **Source:** NIDDK (2023), CDC (2024)

T1D mechanism

Autoimmune destruction of beta cells. The reference lists genetics and environmental triggers such as viruses as risk factors.

T2D mechanism

Muscle, fat, and liver cells respond poorly to insulin. The pancreas initially compensates by producing more insulin; eventually beta cells may not meet demand, causing hyperglycemia.

T2D risk factors

- Overweight or obesity.
- Age 45 years or older.
- Family history of T2D.

- Physical inactivity (less than three times per week, as stated in the source).
- History of gestational diabetes.

What is insulin resistance?

When cells in muscle, fat, and liver do not respond well to insulin.

DM-004: Symptoms and presentation

Topic: Symptoms **Source:** General educational overview; verify against adopted references

Commonly reported symptoms

Increased thirst, frequent urination, increased hunger, unexplained weight loss, fatigue, blurred vision, and slow-healing sores may occur. Some people, particularly with T2D, may have few or no symptoms.

Can symptoms diagnose diabetes?

Symptoms alone do not establish a diagnosis; clinical testing and professional assessment are needed.

DM-005: Diagnosis of diabetes

Topic: Diagnostic criteria **Source:** ADA (2024), educational thresholds

A1C

6.5% or higher is in the diabetes range; A1C estimates average glucose over roughly 2–3 months.

Fasting plasma glucose

126 mg/dL or higher after at least 8 hours without caloric intake is in the diabetes range.

2-hour OGTT

200 mg/dL or higher during a 75-g oral glucose tolerance test is in the diabetes range.

Random plasma glucose

200 mg/dL or higher with classic hyperglycemia symptoms or hyperglycemic crisis can meet a diagnostic criterion.

Caution

In the absence of unequivocal hyperglycemia, confirmation may be needed. Diagnosis should be made by a qualified clinician.

DM-006: Prediabetes

Topic: Diagnostic ranges **Source:** ADA (2024)

A1C

5.7%–6.4%.

Fasting glucose

100–125 mg/dL.

2-hour OGTT

140–199 mg/dL.

Meaning

Glucose is above the usual range but below the diabetes threshold; it indicates increased risk and is not itself a diabetes diagnosis.

DM-007: Monitoring and A1C

Topic: Monitoring **Source:** ADA (2024), general education

What does A1C measure?

It estimates average blood glucose over approximately 2–3 months.

Does A1C show every glucose change?

No. It does not show every fluctuation or an acute low; interpretation depends on clinical context.

DM-008: Self-management

Topic: Care topics **Source:** General educational overview

Common care components

- Taking prescribed medicines as directed.
- Follow-up visits and glucose monitoring when recommended.
- Healthy eating and physical activity suited to the individual.
- Sleep and avoiding tobacco.

Safety

Do not start, stop, or change prescribed medication based on this assistant; consult the treating healthcare team.

DM-009: Nutrition and physical activity

Topic: Lifestyle **Source:** General education

Nutrition

No single meal plan fits everyone. Food and carbohydrate planning should account for health needs, preferences, culture, and prescribed treatment.

Activity

Physical activity can be part of self-management, but an appropriate plan depends on the person and clinical guidance.

DM-010: Medication classes

Topic: Therapeutics **Source:** ADA Standards of Care (2024); limited class overview

Medication selection

Depends on diabetes type, glucose levels, other conditions, risks, preferences, and clinician assessment. The assistant must not choose a medicine for a specific user.

SGLT2 inhibitors

A class that acts through the kidneys to increase urinary glucose excretion. Benefits and indications vary by medicine and clinical context.

Metformin

A commonly used T2D medicine; it primarily reduces hepatic glucose production and can improve insulin sensitivity. Suitability requires clinician review.

Other classes

GLP-1 receptor agonists, DPP-4 inhibitors, and sulfonylureas have distinct mechanisms and safety profiles; this reference does not support individualized drug selection.

DM-011: Insulin overview

Topic: Therapeutics **Source:** ADA Standards of Care (2024), supplied reference

When is insulin required?

The reference states insulin is required for all people with T1D and for some people with T2D experiencing severe hyperglycemia or beta-cell depletion.

Rapid-acting

Lispro/aspart: onset 10–15 min, peak 1–2 h, duration 3–5 h; mealtime coverage.

Short-acting

Regular: onset 30–60 min, peak 2–4 h, duration 5–8 h; the reference states it is injected 30 minutes before a meal.

Intermediate-acting

NPH: onset 1–2 h, peak 4–12 h, duration 14–24 h; often twice daily as basal insulin per reference.

Long-acting

Glargine: onset 1–2 h, relatively flat peak, duration up to 24 h; basal insulin per reference.

Dose safety

Never calculate or recommend an individual insulin dose or adjustment.

DM-012: Hypoglycemia

Topic: Acute glucose event **Source:** ADA Standards of Care (2024), source evaluation query

What is it?

Hypoglycemia means low blood glucose. The source's example asks about a reading of 60 mg/dL; response depends on symptoms, context, and the person's care plan.

Possible symptoms

Shakiness, sweating, hunger, palpitations, dizziness, confusion, or weakness may occur.

Emergency safety

If unconscious, seizing, or unable to swallow safely, call emergency services; do not give food or drink by mouth. If alert, follow the person's clinician-provided plan; do not invent individualized treatment.

DM-013: Hyperglycemia and emergencies

Topic: Emergencies **Source:** ADA Standards of Care (2024), supplied reference

DKA

Most common in T1D in the reference; profound insulin deficiency causes fat breakdown and acidic ketones. Listed features include glucose >250 mg/dL, fruity breath, nausea, Kussmaul respirations, and confusion.

HHS

Most common in older adults with T2D in the reference; severe glucose >600 mg/dL, extreme dehydration, altered mental status, usually without significant ketoacidosis.

Action

Suspected DKA/HHS requires immediate medical attention. The reference describes hospital treatment with IV fluids, insulin, and electrolyte replacement. Do not suggest home management or delay care.

DM-014: Long-term complications

Topic: Complications **Source:** WHO (2023), ADA (2024), supplied reference

Organs and systems

Chronic uncontrolled hyperglycemia can damage the heart, blood vessels, eyes, kidneys, and nerves.

Caution

These are possible complications, not inevitable outcomes; do not estimate personal risk without clinical data.

DM-015: Assistant grounding and safety policy

Topic: RAG behavior **Source:** Project rules

Grounding

Use retrieved passages only; acknowledge missing or incomplete evidence.

Citations

Render section ID and page from retrieved metadata; never fabricate a citation.

Medical boundaries

No diagnosis, prescriptions, individualized dosing, medication changes, or replacing emergency services.

Privacy

Do not request identifying details or unnecessary personal health information.

Question bank for retrieval evaluation

Use these as test queries. Expected IDs are retrieval targets, not answer text. Keep this section out of the indexed knowledge chunks.

- Q01. What is diabetes mellitus? (Expected: DM-001)
- Q02. What does insulin do? (Expected: DM-001)
- Q03. How do type 1 and type 2 diabetes differ? (Expected: DM-002, DM-003)
- Q04. Can T1D begin in adulthood? (Expected: DM-002)
- Q05. What proportion of cases are T2D in this reference? (Expected: DM-002)
- Q06. What happens to beta cells in T1D? (Expected: DM-003)
- Q07. Explain insulin resistance. (Expected: DM-003)
- Q08. Which T2D risk factors are listed? (Expected: DM-003)
- Q09. What is gestational diabetes? (Expected: DM-002)
- Q10. What does A1C measure? (Expected: DM-005, DM-007)
- Q11. What A1C is diabetes range? (Expected: DM-005)
- Q12. What A1C range is prediabetes? (Expected: DM-006)
- Q13. What fasting glucose range is prediabetes? (Expected: DM-006)
- Q14. What are diabetes-range glucose criteria? (Expected: DM-005)
- Q15. Can symptoms alone diagnose diabetes? (Expected: DM-004, DM-005)
- Q16. What symptoms may occur? (Expected: DM-004)
- Q17. What are self-management components? (Expected: DM-008)
- Q18. Is there one diet for everyone? (Expected: DM-009)
- Q19. What do SGLT2 inhibitors do generally? (Expected: DM-010)
- Q20. When does the reference say insulin is required? (Expected: DM-011)
- Q21. Compare insulin types and durations. (Expected: DM-011)
- Q22. Can you calculate my insulin dose? (Expected: DM-011, DM-015)
- Q23. What safety context matters for glucose 60 mg/dL? (Expected: DM-012)
- Q24. What if someone is unconscious with possible hypoglycemia? (Expected: DM-012)
- Q25. What is DKA and what signs are listed? (Expected: DM-013)
- Q26. How is HHS described differently from DKA? (Expected: DM-013)
- Q27. What should happen if DKA/HHS is suspected? (Expected: DM-013)
- Q28. Which organs can chronic hyperglycemia damage? (Expected: DM-001, DM-014)
- Q29. What if retrieved passages do not answer? (Expected: DM-015)
- Q30. Can the assistant choose medication for a user? (Expected: DM-010, DM-015)

Evaluation checklist

- Recall@k: is an expected section among the top-k retrieved?
- MRR: rank of the first expected section.
- Faithfulness: are answer claims supported by retrieved text?
- Citation accuracy: does the cited section/page contain the claim?
- Abstention: does the system admit when evidence is missing?

Recommended indexing workflow

- Split by DM section heading first; then chunk long sections if needed.
- Keep section_id, topic, source, page, and chunk index as metadata.

- Exclude this evaluation question bank from indexed knowledge.
- Avoid chunks mixing unrelated sections.
- Cite from metadata in application code, not model-generated labels.

Source note

This is a newly organized educational reference based on the supplied diabetes knowledge-base excerpts, with additional clearly labeled educational scaffolding and RAG design guidance. Validate all clinical thresholds and treatment details against the authoritative references formally adopted by the project before public deployment.